

When the State Matches: State-Run Dating Apps and Infrastructural Violence

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Introduction

In a bid to boost fertility levels, Iran and Japan have recently launched their own dating apps (de Vries 2024). Philosophy is catching up. Bouke de Vries (2024) defends state-run dating apps on the ground that state incentives align more closely with users' interests than commercial ones: a state need not profit by prolonging searches. Martin Beckstein and de Vries (2025) extend the defense to social engineering: states may use dating apps to pursue demographic goals such as raising below-replacement fertility.

This debate sits on a tension most of us feel. On the one hand, sexual and romantic choice should be free; whom we find attractive is not a matter on which justice issues direct demands. On the other hand, when particular groups become systematically invisible on the platforms through which partners are now found, that looks like an injustice. The first thought pushes toward permitting demographic matchmaking, since no one's choices are forced; the second pushes toward condemning it, since some citizens' prospects are quietly organized away.

I argue that the tension resolves once we see that commercial and state ranking are not the same wrong. When a state-run app embeds demographic objectives in its ranking, it commits a wrong that no commercial platform is positioned to commit: commercial ranking disadvantages users, while state ranking demotes citizens. The state wrongs them not by abusing its power but by exercising it as designed. This wrong, I will argue, is a form of active infrastructural violence.

The main argument runs as follows.

- P1. Dating apps constitute relational infrastructure once they become a principal means of meeting partners. - P2. Harm within that infrastructure consists in patterned non-connection, including the status harm of diminished standing. - P3. Neutrality of treatment constrains states, not firms. - P4. A state ranking that embeds demographic objectives violates neutrality of treatment and realizes, by design, a constituent of social inequality. - C. Therefore a state's demographic ranking commits a kind of wrong no

commercial platform can commit: active infrastructural violence.

Three limits of scope. First, mine is not a comparative claim about magnitudes: whether state apps harm users more than commercial ones do is an empirical question on an axis (benefit) orthogonal to mine (status). Second, it is not an argument for privatization: commercial ranking is objectionable on other grounds. Third, it does not condemn every state-run app: de Vries's original proposal survives my presumption, though, as I argue in the conclusion, only conditionally.

Section 2 develops the infrastructural framework and secures P1 and P2. Section 3 reconstructs and answers the strongest existing defense of demographic matchmaking, securing P3 and P4. Section 4 takes up four objections. Section 5 concludes.

Dating Apps as Relational Infrastructure

2.1 Infrastructural violence

Following Dennis Rodgers and Bruce O'Neill (2012), I define infrastructural violence as serious, systematic harm produced through the organization of an infrastructural network. It comes in two forms. Active infrastructural violence involves articulations of infrastructure "designed to be violent, whether in their implementation or in their functioning" (2012, 406): Haussmann's boulevards, Moses's low overpasses. Passive infrastructural violence arises where harmful effects "derive from infrastructure's limitations and omissions rather than its direct consequences" (2012, 407): disconnection from water networks, exclusion from grids. The concept is deliberately narrower than Galtung-style structural violence: infrastructure is observable and its stakeholders identifiable, a concrete site of responsibility where structural violence offers only an abstraction.

2.2 When, not what

Are dating apps infrastructure? That question is badly formed. As Susan Leigh Star and Karen Ruhleder argue, "infrastructure is a fundamentally relational concept. It becomes infrastructure in relation to organized practices" (1996, 112); the right question is "when—not what—is an infrastructure" (1996, 112). Without a Braille terminal the Internet does not support a blind person's communication (1996, 112): the same arrangement is infrastructure for some and not for others.

Hence my first premise is deliberately conditional: dating apps constitute relational infrastructure once they become a principal means of meeting partners. The condition is not a hedge but a conceptual requirement; on the relational account, no stronger claim is available. And the condition is increasingly satisfied: where network effects make a platform the main gateway to intimate relationships, the alternative routes that would let a user shrug off its ordering disappear.

2.3 Cause, medium, constitution

Infrastructure relates to harm in three ways. First, infrastructure can cause harm: a contaminated water system produces disease. Second, it can mediate harm: existing racial exclusion is transmitted and amplified through transit and housing networks. Third, it can constitute the field of status and possibility itself: a visibility ranking partly makes it the case who counts as a desirable partner.

Dating apps do all three, but the argument's weight rests on constitution. Causal and mediating relations can be handled with familiar vocabularies of distributive justice and causal responsibility: there is a prior good to distribute, or a prior injustice transmitted. Constitution is different. Before the ranking operates, the status "visible on this app" does not exist; the ranking brings the status system into being. So the question "was the distribution fair?" arrives too late. The prior question is whether this status system may be created at all, and that question demands the vocabulary of relational equality, not of distribution.

2.4 The harm: patterned non-connection

What is the harm at this level? Not that any user is prevented from meeting any particular person; rather, particular groups come to appear systematically less often in one another's option sets: patterned non-connection. This includes a status harm. On Elizabeth Anderson's relational account, equality is not a pattern of distribution but a social relation, and what egalitarians have historically opposed are relations of superiority and inferiority among citizens (1999, 313). A network ordering that assigns some citizens diminished standing in the visibility order is a harm of this relational kind, whatever its aggregate welfare effects.

This harm is structurally hard to see. Infrastructure normally becomes visible on breakdown (Star and Ruhleder 1996, 113), but an absent match does not look like a breakdown; it looks like the system working normally. And each user experiences persistent non-matching as personal bad luck rather than structural output, so no collective grievance ever forms. This invisibility will matter in Section 3: it is why the wrong escapes frameworks that look for harm at the point of individual decision.

The Neutrality Argument

3.1 The social-engineering defense

Beckstein and de Vries's case for demographic matchmaking rests on three claims, which I state in their strongest form. (S1) Fertility-oriented design serves rather than overrides users' preferences: family-goal filters benefit even those who want no children, since they are spared incompatible matches. (S2) Liberal states already accept policies that bear unequally on citizens' conceptions of the good, such as child benefits,

incest prohibitions, and default-based nudges, so demographic matchmaking extends a recognized practice. (S3) The interventions coerce no one: “no one is being prevented from dating people of their own choosing” (2025, 9).

Against S1, the authors’ own concession is decisive. They grant that algorithms may be skewed toward fertility-promoting matches “even without explicit users’ consent” (2025, 9). When the state’s demographic objective conflicts with users’ sorting preferences, the objective wins. A system of which that is true does not merely serve preferences; it structures their exercise. The preference-serving frame does not describe the design but conceals it.

Answering S3 requires no new resources: it locates harm at the point of individual choice, and patterned non-connection (Section 2.4) is precisely the harm that survives that test; no one is prevented from dating anyone, yet groups are ordered out of one another’s option sets.

3.2 What the precedent claim really is

S2 is the defense’s strongest move, and answering it is this paper’s central task. The precedent claim is a move in the liberal neutrality debate, and identifying which move defeats it.

The observation that liberal states “already accept policies bearing unequally on citizens’ conceptions of the good” is an observation that a certain standard, neutrality of effects, is routinely violated. But as Alan Patten (2012) shows, neutrality of effects is a standard nobody holds: under it virtually every policy comes out non-neutral, since even the legal protection of the basic liberties disadvantages unpopular ways of life, and liberals “do not recognize a general duty to be neutral with respect to effects” (Patten 2012). The failure of a standard no one endorses justifies nothing.

The applicable standard is what Patten calls neutrality of treatment, which a state violates when its policies are “more accommodating, or less accommodating, of some conceptions of the good than they are of others” (2012). Where neutrality of effects equalizes across policy outputs, neutrality of treatment equalizes across the state’s inputs. By this standard, child benefits, like police and fire protection, accommodate rival conceptions of the good on equal terms; they are cases of what Patten calls generic entanglement, state involvement with goods that nearly all conceptions value. The precedents are precedents of neutrality observed, not of neutrality breached.

A fertility weighting is different in kind. Consider Patten’s tax case: a state that taxes the goods used by one conception of the good at 15% and those used by a rival conception at 25% is less accommodating of the rival, even if it aims at no one’s conversion and changes nothing about either conception’s popularity (2012). A ranking parameter that boosts visibility for fertility-compatible profiles is the algorithmic counterpart

of that differential tax: it taxes the good used by child-free, queer, and older citizens' conceptions of intimate life (visibility) at a higher rate than the pronatalist conception's. The weighting is an explicit parameter of the policy, so the unequal input cannot hide behind the generality of the service.

One can identify and decline a child benefit; one cannot refuse, or even detect, one's own deranking. But that asymmetry is auxiliary: it concerns the observability of the harm, and whether the state's treatment is neutral is settled independently of it.

3.3 Why the state

Everything so far indicts demographic ranking as such; what makes the state distinctive? Three independent grounds.

First, neutrality of treatment is a constraint on states, not firms. A commercial platform is not bound by liberal neutrality and so is not positioned to violate it; the wrong of commercial ranking, real as it may be, is a wrong of a different kind, to be criticized on other grounds.

Second, on Niko Kolodny's account, social inequality partly consists in some having "attributes (for example, race, lineage, wealth, perceived divine favor) that generally attract greater consideration than the corresponding attributes of others" (2014, 296). A state that allocates visibility by fertility makes such an attribute, and the age, health, and sexual orientation correlated with it, generally consideration-attracting by design. This component does not ask how the greater consideration comes about; ranking produces the required generality, and produces it institutionally. Moreover, Kolodny's first component requires that greater power be accompanied by a resolute disposition to refrain from exercising it (2014, 295); a state that embeds the objective in the ranking is not merely undisposed to refrain but exercising the power deliberately. This is the sense in which the state wrongs citizens not by abusing its power but by exercising it as designed. Since, for Kolodny, we have reason to avoid such relations "for their own sake, not simply as a symbol of, or means to, something else" (2014, 300), I need not separately show that the ranking expresses disrespect or produces bad outcomes.

Third, Anderson. What citizenship secures includes standing as an equal in civil society, whose institutions expressly include communications systems (1999, 317). In a state-run matching platform, the body charged with guaranteeing that standing is the very body ranking it. No commercial platform can instantiate this reflexive wrong, because no firm bears the guarantor's obligation in the first place.

With P3 and P4 secured, the conclusion follows. A state's demographic ranking demotes citizens' standing by design, a wrong of a kind commercial ranking cannot commit; in the terms of Section 2.1, it is active infrastructural violence.

Objections

4.1 No one has a claim to sexual attention

The strongest objection attacks P2. Following Amia Srinivasan, Elsa Kugelberg holds that because of the strong value of choice in the sexual sphere, there can be no right to sex (Kugelberg 2025). No one owes anyone sexual attention. If so, reduced visibility violates no right, and the complaint at the heart of my argument has no ground.

The objection is right about what it asserts and wrong about what it targets. Anderson's distinction does the work: "Although the distribution of natural assets is not a matter of justice, what people do in response to this distribution is" (1999, 331). The natural distribution of attractiveness generates no claims to compensation, and no one may be required to find anyone attractive. But institutions may not make the possession of a disability, an unattractive appearance, or, I add, low expected fertility "the occasion" for exclusion from standing in civil society (1999, 331). My argument demands no one's attention. It demands that the state not make fertility the occasion for ranking citizens. The absence of a claim-right underwriting the first does nothing to weaken the second.

4.2 Matching is generic entanglement

The defender of demographic matchmaking has a strong reply to Section 3.2: matching services, like police and fire protection, are goods nearly every conception of the good values. If the state provides matching to all, it engages in generic entanglement, which by my own Pattenite standard satisfies neutrality of treatment.

Grant the premise: a matching service as such is generic entanglement. The objection fails because it conflates the service with the weighting. A fire service that responds to every building, but is designed to reach church fires first, does not have its priority rule laundered by the generality of the service. Likewise, the fertility weighting is a device within the service that taxes rival conceptions' goods at different rates, and it is the weighting, not the service, that my argument condemns.

A fallback presses formal generality: the design favors no citizen, only a conception, and anyone can adopt the pronatalist conception and enjoy the favor. Patten's lacrosse case answers this in advance. A policy funding only lacrosse facilities is formally open to everyone, since any citizen may take up lacrosse, yet it departs from neutrality of treatment: the unfairness consists in its conflict with non-beneficiaries' interest in pursuing the conception of the good they actually hold (Patten 2012). Telling child-free citizens they would be favored if they converted concedes the violation of that interest; it does not deny it.

4.3 This is just the manipulation objection

One might ask whether my critique is an instance of the existing critique of automated influence, making the infrastructural framework idle. Claire Benn and Seth Lazar (2022) give the most sophisticated version of that critique, and they too favor structural over interactional evaluation, so the suspicion is natural.

But their manipulation concept targets a different level. Manipulation, for them, involves undermining an individual's decision-making power, "their ability to select among their options, given their beliefs about the world, in ways that advance their goals" (2022, 139). Deranking undermines no such ability. The user confronting her feed chooses among the presented profiles with her rational capacities fully intact; what has been shaped is which profiles stand before her. My objection targets the constitution of the option set, a level their formulation does not address and one that must be prior: choosing among one's options presupposes that the option set was not composed against one's standing. What the two critiques share is the structural method, not the argument. Indeed Benn and Lazar independently oppose giving states this kind of access, on power-balance grounds (2022, 145); my narrower, status-based case leaves de Vries's original proposal standing where theirs would sweep it away, and that difference in extension confirms the difference in ground.

4.4 Population collapse defeats the presumption

Patten's neutrality is a *pro tanto* constraint, not a prohibition, and the objector may urge that sub-replacement fertility is weighty enough to defeat it. The reply applies Patten's own test: a departure is justified only if the countervailing consideration is weighty and can be advanced through, and only through, neutrality-curtailling measures (Patten 2012). Fertility can be pursued through childcare provision, housing policy, and income transfers. The second condition therefore fails: to defeat the presumption, the defender of demographic ranking must show that no neutrality-respecting means would do. Nothing in the current defense attempts that showing.

Conclusion

The argument splits the two defenses with which I began. What does not survive is a ranking system encoding collective demographic ends: it violates neutrality of treatment, realizes a constituent of social inequality by design, and enacts the reflexive wrong of a guarantor ranking the standing it must guarantee. De Vries's original proposal, the state app that merely declines to profit from prolonged searches, survives the presumption, but only conditionally. Its ranking must be objective-free, yet "apps cannot help but intervene—they need to sort profiles in some order. The option to do nothing does not exist" (Kugelberg 2025), and any sorting rule is more accommodating of some conceptions of intimate life than of others. And the state must be resolutely

disposed to refrain from demographic use, a condition disconfirmed by every existing state app, launched precisely “in a bid to boost fertility levels” (de Vries 2024), and by this literature itself, which moved from neutral instrument to fertility engineering within a year. The concession stands; the class it names may be empty.

The paper’s contribution is twofold. It brings the concept of infrastructural violence into the ethics of intimate technology, giving a name and a normative structure to harms that decision-level frameworks cannot register. And it shifts the debate’s question: no longer whether state-run dating apps can be made transparent and abuse-resistant, but whether the state may organize citizens’ intimate visibility around collective ends at all.

References

- Anderson, E. S. (1999). What Is the Point of Equality? *Ethics*, 109(2), 287-337. - Beckstein, M., & de Vries, B. (2025). Dating apps as tools for social engineering. *Ethics and Information Technology*, 27, 11. - Benn, C., & Lazar, S. (2022). What’s Wrong with Automated Influence. *Canadian Journal of Philosophy*, 52(1), 125-148. - de Vries, B. (2024). State-run dating apps: Are they morally desirable? *Philosophy & Technology*, 37, 30. - Kolodny, N. (2014). Rule Over None II: Social Equality and the Justification of Democracy. *Philosophy & Public Affairs*, 42(4), 287-336. - Kugelberg, E. (2025). Dating Apps and the Digital Sexual Sphere. *American Political Science Review*. - Patten, A. (2012). Liberal Neutrality: A Reinterpretation and Defense. *Journal of Political Philosophy*, 20(3), 249-272. - Rodgers, D., & O’Neill, B. (2012). Infrastructural violence: Introduction to the special issue. *Ethnography*, 13(4), 401-412. - Srinivasan, A. (2021). *The Right to Sex*. Bloomsbury. - Star, S. L., & Ruhleder, K. (1996). Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces. *Information Systems Research*, 7(1), 111-134.